6th SEMESTER LAB REPORT

TITLE : Introduction to Proteus Software: Simulation and Circuit Elements Placement



Faiq Ali (249520)

Muhammad Zeeshan (243346)

Muhammad Waqas (240756)

BACHELOR OF COMPUTER ENGINEERING (Fall-2025)

Submitted To:

Engr. Shanzay amil

Lab Engineer

DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
FACULTY OF ENGINEERING AIR UNIVERSITY,ISLAMABAD

	AIR UNIVERSITY
	DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
	EXPERIMENT NO 3

Lab Title: Introduction to Proteus Software: Simulation and Circuit Elements Placement

Objective: 1. Introduction to Proteus Software: Simulation and Circuit Elements Placement

Student Name: **Muhammad Waqas -Muhammad Zeesha -Faiq Ali**

LAB ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Ability to Conduct Experiment					
Ability to assimilate the results					
Effective use of lab equipment and follows the lab safety rules					

Total Marks: _____ Obtained Marks: _____

LAB REPORT ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Data presentation					
Experimental results					
Conclusion					

Total Marks: _____ Obtained Marks:

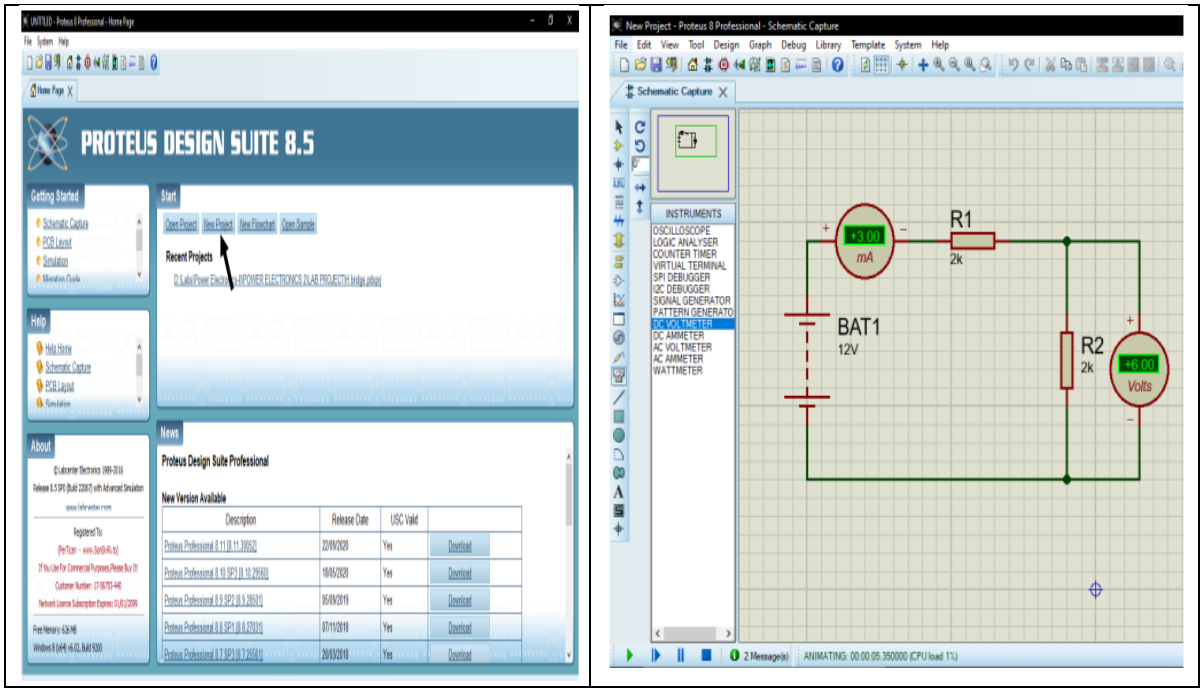
Date: 25/March/2025 Signature:

LABORATORY EXPERIMENT NO. 6

INTRODUCTION TO PROTEUS SOFTWARE: SIMULATION AND CIRCUIT ELEMENTS PLACEMENT

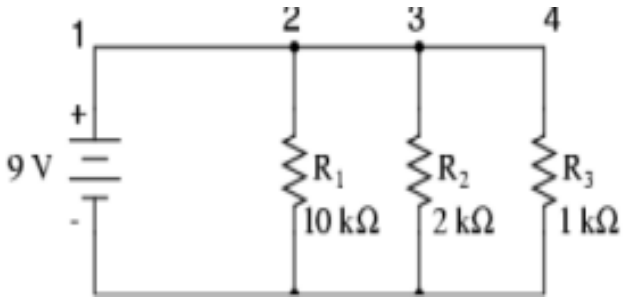
OBJECTIVE:

1. To learn the use of simulation software for circuit analysis.
2. To implement circuits in PROTEUS for comparison purposes



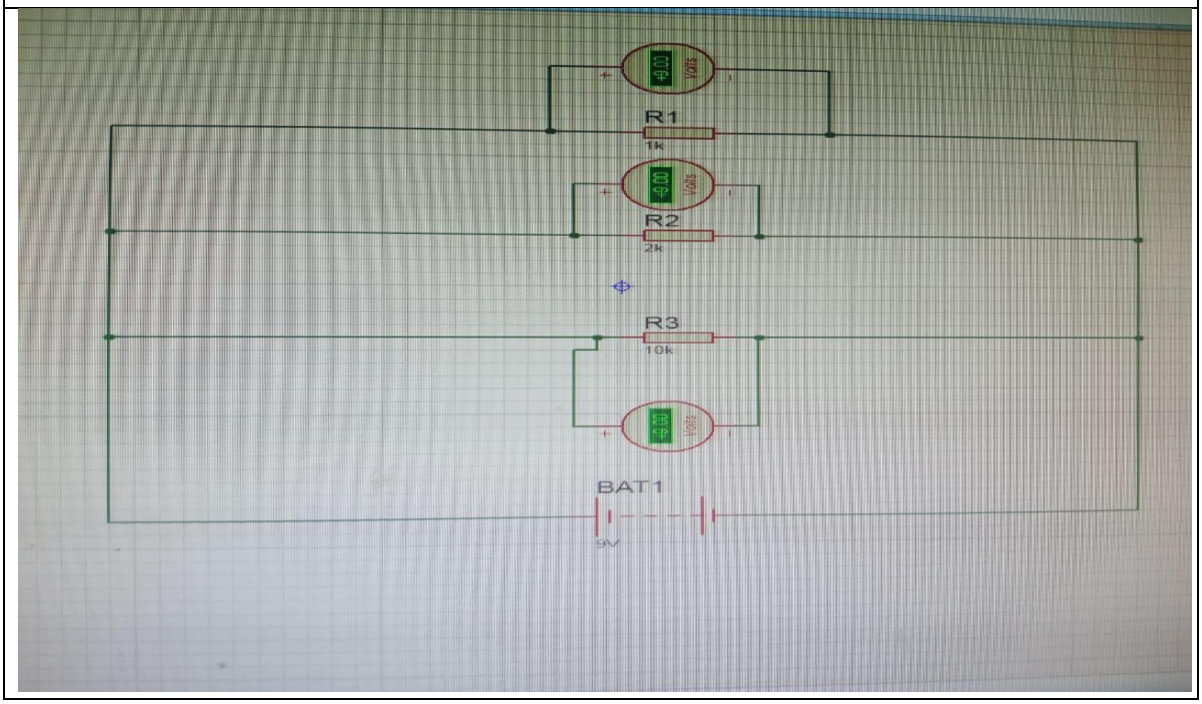
The Proteus schematic capture program is an experimental canvas for students. Placing and wiring is very intuitive and with tens of thousands of components to simulate.curiosity and creativity can be encouraged in equal measure.

a) Simulate to measure the current flowing through and voltage across each resistor of following circuits and paste your results in the empty box below.

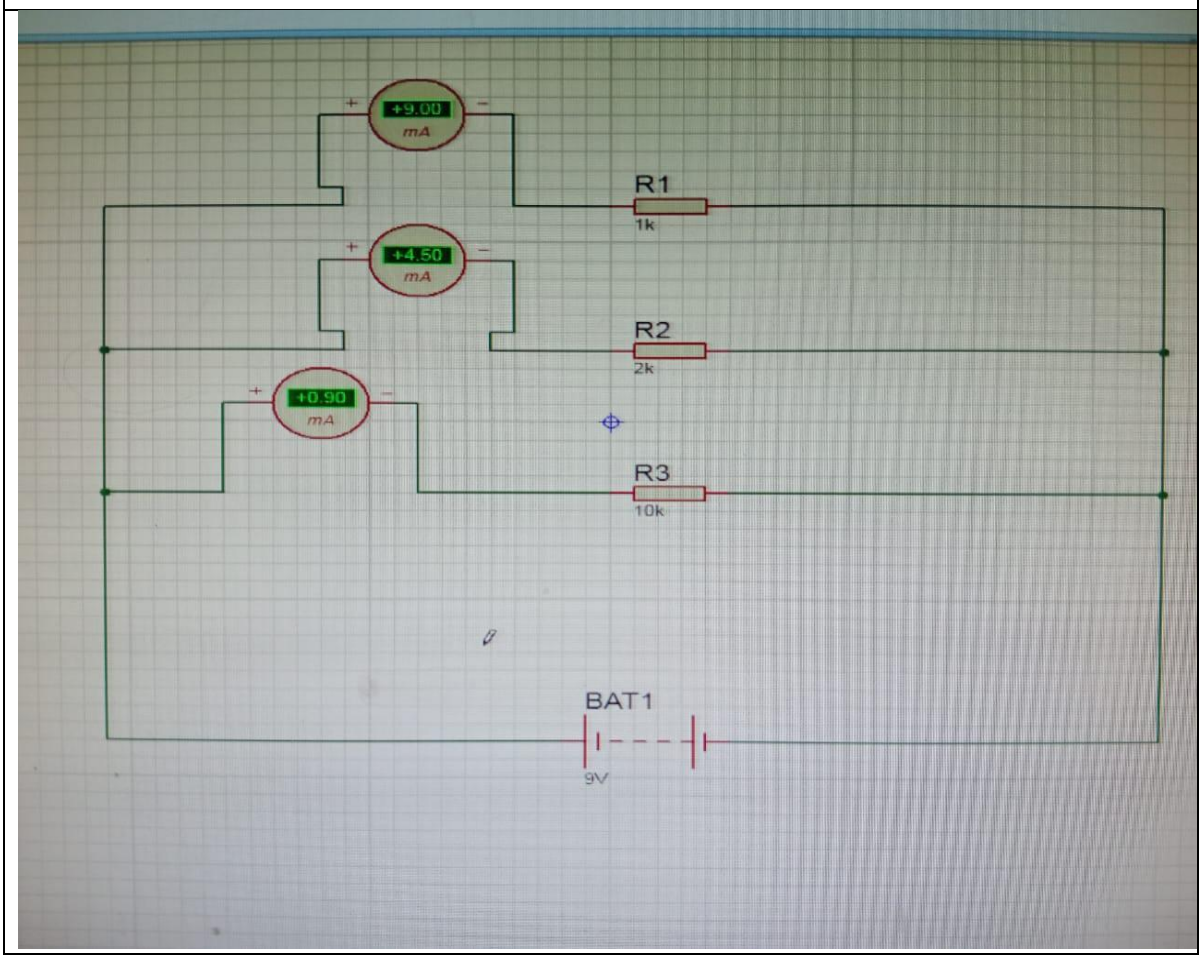


Resstors	Current	Voltages
R1=1000 ohm	+9.00mA	+9.00V
R2=2000 ohm	+4.00mA	+9.00V
R3 =10000 ohm	+0.90mA	+9.00V

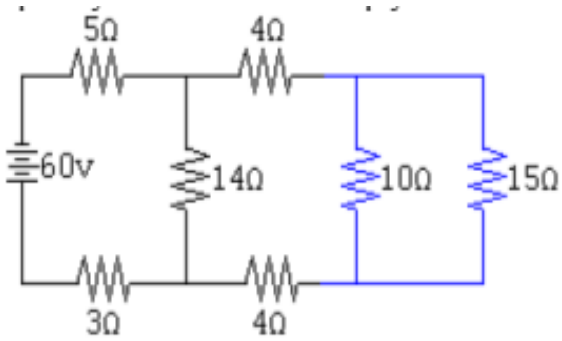
Voltmeters are in parallel to measure voltages across each resistors



Ammeters are in seres to measure voltages across each resistors

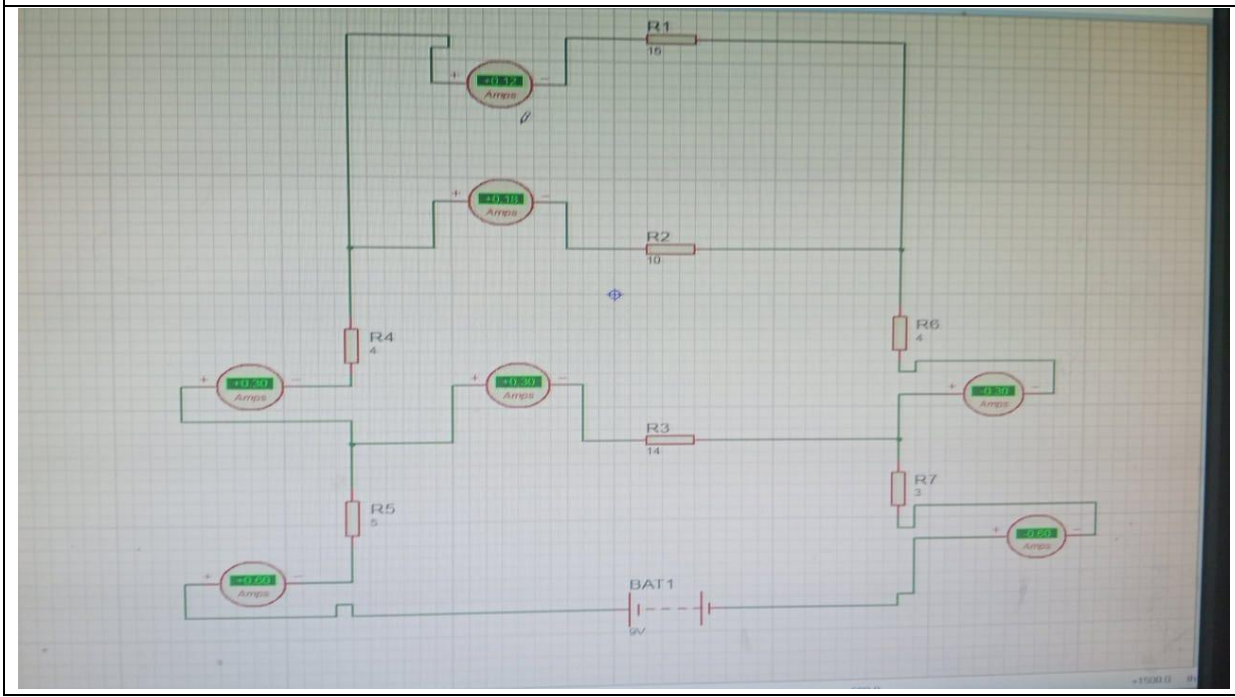


b) Simulate to measure the current flowing through and voltage across each

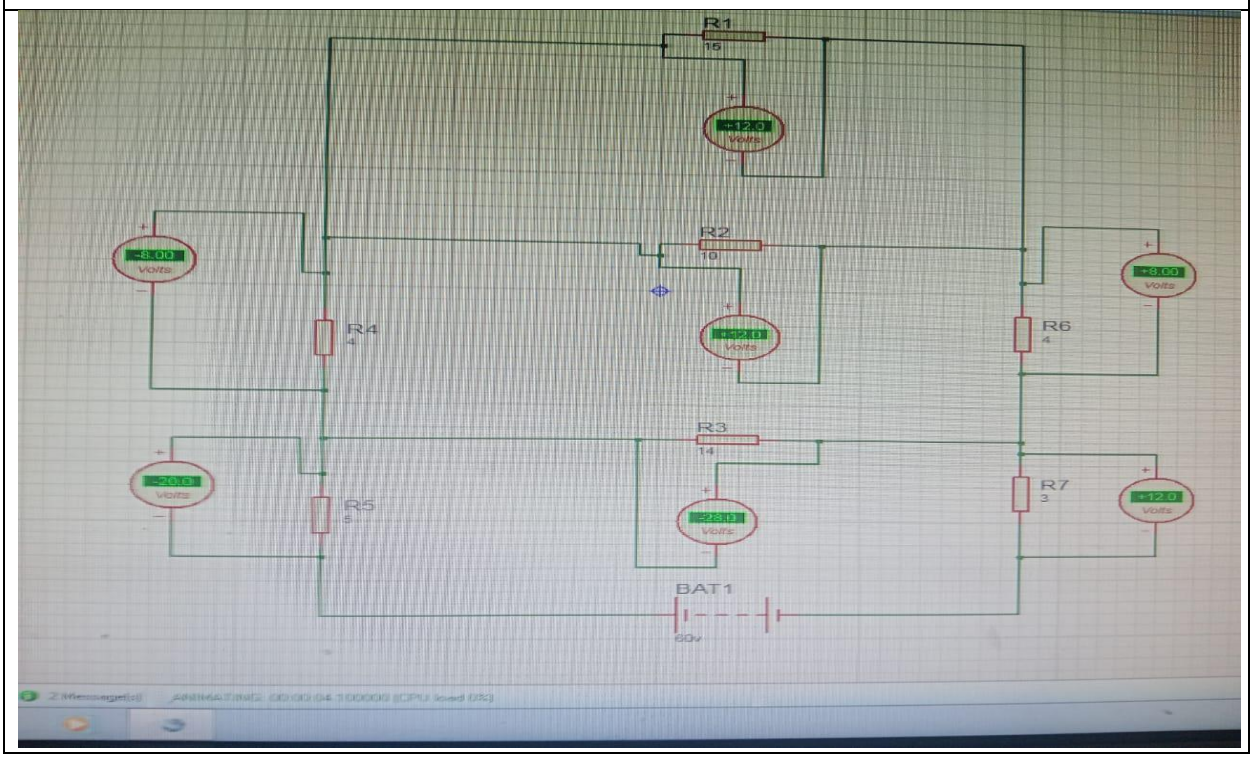


Resistors	Voltages	Current
R1=15 ohm		+0.12 A
R2=10 ohm		+0.18 A
R3=14 ohm		0.30 A
R4=4 ohm		+0.30A
R5=5 ohm		+0.50 A
R6=4 ohm		+0.30 A
R7=3 ohm		+0.60 A

Ammeters are in in series across each resistor to measure current



Volteters are in in parallel across each resistor to measure voltages



Conclusion:

The Proteus schematic capture program offers students a dynamic space for experimentation. Its intuitive interface makes placing and connecting components simple, while a vast library of simulated parts encourages both curiosity and creativity. Combined with its powerful mixed-mode SPICE simulation engine, Proteus creates a safe, efficient, and immersive learning experience.Students can interact with live simulations by pressing buttons, adjusting potentiometers, or toggling switches, making it an engaging tool for understanding electronics. At beginner levels, visual indicators for voltage and current help in grasping basic concepts. As they progress, students can use meters for measurements and explore advanced tools like oscilloscopes and logic analyzers for deeper circuit analysis. For those at higher levels, Proteus enables the study of complex circuits with detailed graphical analyses, including frequency response, Fourier transformations, and distortion measurements.